RESOLUTION MSC.565(108) (adopted on 24 May 2024)

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necessary with regard to ignitability of hydrogen.
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4.10.1.2 Furthermore, with regard to the wide range of flammable limits of hydrogen, the increased quantities of carbon dioxide as a fire -extinguishing medium should be specified as mentioned in paragraph 4.7. No additional special requirement is considered to be necessary with regard to the wide range of flammable limits of hydrogen.
4.10.2 Disposal of cold hydrogen gas
The wide flammability range makes disposal of cold hydrogen gas a major hazard. Cold plumes of released hydrogen may impede adequate dilution of hydrogen down to below 4% and may lead to flash-back to the vent from distant ignition sources outside safety -controlled areas. The low ignition energy and wide flammable range may present significant challenges.

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areas. The low ignition energy and wide flammable range may present significant challenges.
4.11 Prevention of dangerous purging operation
4.11.1 During cargo operations for maintenance, pipes and tanks should be purged with an inert gas or inert gases as illustrated in the figure below. For safety, due consideration should be given to temperature and boiling points of the inert gases. Residual pockets of hydrogen or the purge gas will remain in the enclosure if the purging rate, duration, or extent of mixing is too low. Therefore, reliable gas concentration measurements should be obtained at a number of different locations within the system for suitable purges. Temperature should also be measured at a number of locations. Oxidizing agents may exist in a hydrogen containing equipment, specifically: air, cold box atmospheres containing air diluted with nitrogen, or oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.

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oxygen-enriched air that can be condensed on process pipe work within the cold box in special circumstances.
4.11.2 There are special measures that may need to be put in place in order to mitigate the hazards, e.g. air should be eliminated by nitrogen purge prior to introduction of hydrogen into cargo piping or processing equipment. Nitrogen should then be eliminated by hydrogen purge, where there is a possibility of its solidification in the subsequent process.
Figure 1
References in part A
1 ISO/TR 15916:2015, Basic consideration for the safety of hydrogen systems (ISO)
2 American Institute of Aeronautics and Astronautics, "Safety Standard for Hydrogen

and Hydrogen Systems (Guide to Safety of Hydrogen and Hydrogen Systems)", 2005
(AIAA)
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ANNEX 20
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THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
NOTING that the International Convention for the Safety of Life at Sea ("the Convention"), 1974 and the International Code for the Construction and Equipment of Ships Carrying Liquefied Gases in Bulk ("the IGC Code") currently do not specifically provide requirements for carriage of liquefied hydrogen in bulk by sea,
NOTING ALSO that paragraph 5 of Preamble of the IGC Code states that requirements for new products and their conditions of carriage will be circulated as recommendations, on an
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new products and their conditions of carriage will be circulated as recommendations, on an interim basis, prior to the entry into force of the appropriate amendments,
RECOGNIZING a need for the development of the Revised interim recommendations for carriage of liquefied hydrogen in bulk,
ACKNOWLEDGING that, in the interim, there is an urgent need to provide recommendations to the Administrations on safe carriage of liquefied hydrogen in bulk,
ACKNOWLEDGING ALSO that the Revised Interim Recommendations are intended to facilitate establishment of a tripartite agreement for a pilot ship, which will be developed for the research and demonstration of safe long-distance overseas carriage of liquefied hydrogen in bulk,
HAVING CONSIDERED the Revised Interim Recommendations prepared by the Sub - Committee on Carriage of Cargoes and Containers at its ninth session,
1 ADOPTS the Revised interim recommendations for carriage of liquefied hydrogen in
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1 ADOPTS the Revised interim recommendations for carriage of liquefied hydrogen in bulk, the text of which is set out in the annex to the present resolution;
2 INVITES Member States to apply the Revised Interim Recommendations to the pilot ship carrying liquefied hydrogen in bulk taking the explanatory notes into consideration;
3 AGREES to acquire information on safe carriage of liquefied hydrogen in bulk prior to amendment to the IGC Code for the inclusion of liquefied hydrogen;
4 ALSO AGREES that these Revised Interim Recommendations may need to be reviewed if they are to be applied to ships other than the pilot ship; and
5 URGES Member States and the industry to submit information, observations, comments and recommendations based on the practical experience gained through the application of the Revised Interim Recommendations and submit relevant safety analysis on ships carrying liquefied hydrogen in bulk.
6 REVOKES resolution MSC.420(97).

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on ships carrying liquefied hydrogen in bulk.
6 REVOKES resolution MSC.420(97).
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ANNEX
REVISED INTERIM RECOMMENDATIONS FOR CARRIAGE
OF LIQUEFIED HYDROGEN IN BULK
1 INTRODUCTION
1.1 For the carriage of liquefied gases in bulk by ships, the ships should comply with the relevant requirements in the IGC Code, as amended ("the Code"). The scope of the Code provided in paragraph 1.1.1 is:
"The Code applies to ships regardless of their size, including those of less than 500 gross tonnage, engaged in the carriage of liquefied gases having a vapour pressure exceeding 0.28 MPa absolute at a temperature of 37.8°C, and other products, as shown in chapter 19, when carried in bulk".
1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen
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products, as shown in chapter 19, when carried in bulk".
1.2 A ship carrying liquefied hydrogen in bulk (hereinafter called "liquefied hydrogen carrier") should comply with the Code.
1.3 The Code requires that a gas carrier should comply with the minimum requirements for the cargo listed in chapter 19. However, the requirements for liquefied hydrogen are not specified in the Code.
1.4 This annex provides the Revised Interim Recommendations, as referred to in paragraph 5 of the preamble of the Code, for the carriage of liquefied hydrogen in bulk, which are intended to provide the basis for the future minimum requirements for the carriage of this cargo. The Revised Interim Recommendations are intended to facilitate the establishme nt of a tripartite agreement among the relevant Administrations for the carriage of liquefied hydrogen

in bulk. However, they are not intended to prohibit the adoption of designs and arrangements
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in bulk. However, they are not intended to prohibit the adoption of designs and arrangements
other than those specified in the Code or in these Recommendations, at the discretion of the Administrations.
1.5 These recommendations have been developed under the assumption that a liquefied hydrogen carrier does not carry liquefied gases other than liquefied hydrogen. These recommendations, therefore, are not applicable to liquefied hydrogen carriers carrying gase s other than liquefied hydrogen.
1.6 In the Code, reference is made to paragraph 5 of the Preamble; paragraph 1.1.6.1; and Note No.8 on completion of certificate in "model form of international certificate of fitness for the carriage of liquefied gases in bulk" in appendix 2 to the Code.
1.7 These Revised Interim Recommendations consist of the following parts. Part A is applicable to ships with any type of cargo containment system. Part B and subsequent part(s)

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applicable to ships with any type of cargo containment system. Part B and subsequent part(s)
prescribe additional special requirements for cargo containment systems of specific types.
Part A: General (applicable to ships with any type of cargo containment system);
Part B: Cargo containment systems of independent cargo tanks using vacuum insulation; and
Part C: Cargo containment systems of independent cargo tanks using insulation materials and hydrogen gas in the inner insulation spaces.
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1.8 Part A of this document was developed based on the design of parts B and C. If subsequent part(s) are added, the special requirements prescribed in part A may be reviewed.
Part A
General
(Applicable to ships with any type of cargo containment system)